Tutorial Sheet-6

Indian Institute of Technology Jodhpur
MAL1020 (Mathematics-II)
Instructors: Dr. Puneet Sharma and Dr. Dilpreet Kaur

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1. Solve the following:
   (a) $y' = (y+4x)^2$
   (b) $e^x y' = 2(x+1)y^2, \quad y(0) = 16$
   (c) $y' = \frac{1-2y-4x}{1+y+2x}, \quad$ (Hint : $y + 2x = v$)
   (d) $2xydx + 3x^2dy = 0$
   (e) $-yx^{-2}dx + x^{-1}dy = 0$
   (f) $\left(x^2-1\right)\frac{dy}{dx} + 2y = (x+1)^2$
   (g) $x\frac{dy}{dx} + (3x+1)y = e^{-3x}$

2. Under what conditions $(ax+by)dx + (kx+ly)dy = 0$ is exact ?

3. Solve the following separable equations:
   (a) $(x-4)y^4dx - x^3\left(y^2-3\right)dy = 0;$
   (b) $x\sin ydx + \left(x^2+1\right)\cos ydy = 0;$
   (c) $y' = \frac{x^2+y^2}{xy} \qquad$ (Substitute $\frac{y}{x} = u$);
   (d) $y' + \operatorname{cosec} y = 0;$
   (e) $\left(x^2-3y^2\right)dx + 2xydy = 0.$

4. Determine whether or not each of the given equations is exact; solve those which are exact.
   (a) $\left(6xy+2y^2-5\right)dx + \left(3x^2+4xy-6\right)dy = 0;$
   (b) $\left(y^2+1\right)\cos xdx + 2y\sin xdy = 0;$
   (c) $(2xy+1)dx + \left(x^2+4y\right)dy = 0;$
   (d) $\left(3x^2y+2\right)dx - \left(x^3+y\right)dy = 0;$
   (e) $-2xy\sin\left(x^2\right)dx + \cos\left(x^2\right)dy;$
   (f) $\left(e^{(x+y)}-y\right)dx + \left(xe^{(x+y)}+1\right)dy.$

5. Determine the constant A such that the equation is exact, and solve the resulting exact equation:
   (a) $\left(x^2+3xy\right)dx + \left(Ax^2+4y\right)dy = 0;$
   (b) $\left(\frac{1}{x^2}+\frac{1}{x^4}\right)dx + \left(\frac{Ax+1}{y^3}\right)dy = 0.$

6. Determine the most general function $N(x,y)$ such that the equation is exact:
   (a) $\left(x^2+x^y2\right)dx + N(x,y)dy = 0;$
   (b) $\left(x^{-2}y^{-2}+xy^{-3}\right)dx + N(x,y)dy = 0.$

7. Solve the following ODEs by method of integrating factors:
   (a) $\left(2y^2+3x\right)dx + 2xydy = 0,$
   (b) $\cos xdx + \left(1+\frac{2}{y}\right)\sin xdy = 0.$

8. Solve:
(a) $\frac{dx}{dy} = \frac{y^2}{1-3xy}$
(b) $\left(6xy + 2y^2 - 5\right) dx + \left(3x^2 + 4xy - 6\right) dy = 0$
(c) $\left(\theta^2 + 1\right) \cos r dr + 2\theta \sin r d\theta = 0$
(d) $\left(ye^x + 2e^x + y^2\right) dx + \left(e^x + 2xy\right) dy = 0, \quad y(0) = 6$
(e) $\left(\cos^2 x - y \cos x\right) dx - (1 + \sin x) dy = 0$
(f) $x^2 y' - 3xy - 2y^2 = 0$
(g) $x \sin\left(\frac{y}{x}\right) \frac{dy}{dx} = y \sin\left(\frac{y}{x}\right) + x$
(h) $\left(y \cos x + 2xe^y\right) dx + \left(\sin x + x^2 e^y - 1\right) dy = 0$
(i) $\left(x^2 - 4xy - 2y^2\right) dx + \left(y^2 - 4xy - 2x^2\right) dy = 0$
(j) $\left(x^3 + xy^3\right) dx + 3y^2 dy = 0$
(k) $(x+1)\frac{dy}{dx} + 1 = 2\exp(-y)$
(1) $\frac{dy}{dx} = \frac{y}{x} + \sin\left(\frac{y}{x}\right)$
(m) $\left(1 + 2xy \cos\left(x^2\right) - 2xy\right) dx + \left(\sin\left(x^2\right) - x^2\right) dy = 0$
(n) $\left(xy^2 - \exp\left(\frac{1}{x^3}\right)\right) dx - x^2 y dy = 0$
(o) $y\left(xy + 2x^2y^3\right) dx + x\left(xy - x^2y^2\right) dy = 0$

9. Are the following set of functions defined on $-\infty < x < \infty$ linearly dependent or independent there? Why?
(a) $\phi_1(x) = 1, \phi_2(x) = x, \phi_3(x) = x^3$;
(b) $\phi_1(x) = x, \phi_2(x) = e^{2x}, \phi_3(x) = |x|$.

10. Consider a differential equation of the form

$$\left[y + xf\left(x^2 + y^2\right)\right] dx + \left[yf\left(x^2 + y^2\right) - x\right] dy = 0$$

(a) Show that an equation of this form is not exact.
(b) Show that $\frac{1}{x^2+y^2}$ is an integrating factor of an equation of this form.